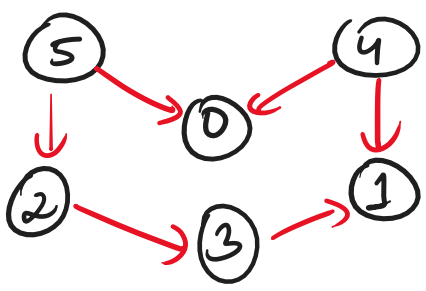


Topological Sort (BFS: Kahn's Algorithm)

$u \rightarrow v$ (linear Ordering)
(u comes before v)



Step 1: Insert all nodes with indegree 0 into queue.
Step 2: Take out node & reduce indegree of neighbours.

1 X
3 X
2 X
0 X
5 X
4 X
queue

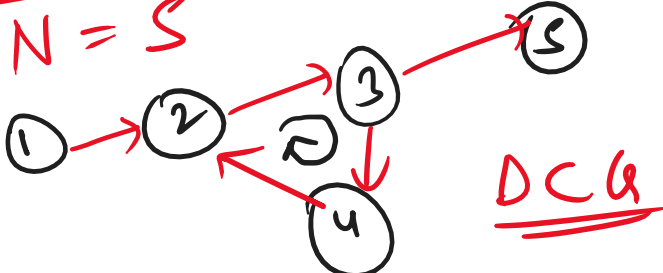
Adj List

0 $\rightarrow$
1 $\rightarrow$
2 $\rightarrow$ 3
3 $\rightarrow$ 1
4 $\rightarrow$ 0, 1
5 $\rightarrow$ 0, 2

New Concept: Indegree $\rightarrow$ No of incoming nodes to a particular node.
O/p: $\rightarrow$ 4 5 0 2 3 1

Cycle Detection using Topo Sort Kahn's Algorithm

(DAG) (DCG) (Directed Cyclic Graph)



Indegree Array: $\rightarrow$ [0, 2, 1, 1, 1]

1
2
3
4
5
queue

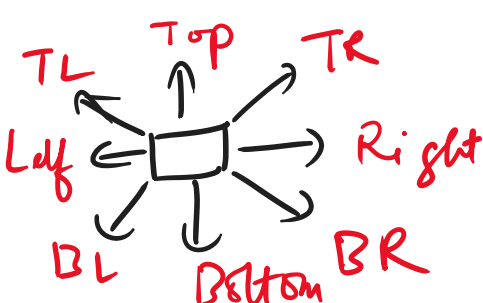
Adj List

1 $\rightarrow$ 2
2 $\rightarrow$ 3
3 $\rightarrow$ 4, 5
4 $\rightarrow$ 2
5 $\rightarrow$

R1: $\Rightarrow$ Topo Sort Output = 1
R2: If the topo sort size == N (false)
size() != N (true) $\Rightarrow$ NOT true Cycle

Number of Islands: $\rightarrow$ GFG (Medium)

L	L	W	W	W
W	L	W	W	L
L	W	W	L	L
W	W	W	W	W
L	W	L	L	W



* Intuition or Approach: $\rightarrow$

* if the current cell is a Land 'L' we will perform a dfs call & try to move in all 8 possible directions.

* The moment a dfs finishes we will track the count

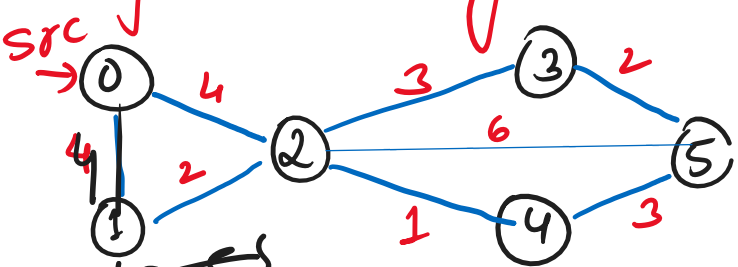
* When all dfs calls are done, we have got the no of islands.

Directions	Change in Coordinates	dx	dy
Top-Left	(x-1, y-1)	-1	-1
Top	(x-1, y)	-1	0
Top-Right	(x-1, y+1)	-1	+1
Left	(x, y-1)	0	-1
Right	(x, y+1)	0	+1
Bottom-Left	(x+1, y-1)	+1	-1
Bottom	(x+1, y)	+1	0
Bottom-Right	(x+1, y+1)	+1	+1

(x, y) No of DFS calls = No of Islands

Shortest distance Algorithms (Single Source) $\rightarrow$ S, PQ, Q

Dijkstra's Algorithm (One source to all other nodes)



* (Better distance) $\rightarrow$ Relaxation

{n, w} Adjacency List

$\rightarrow$ 0 $\rightarrow$ {1, 4} {2, 4}
 $\rightarrow$ 1 $\rightarrow$ {0, 4} {2, 2}
 $\rightarrow$ 2 $\rightarrow$ {0, 4} {1, 2} {3, 3} {4, 1} {5, 3}
 $\rightarrow$ 3 $\rightarrow$ {2, 3} {5, 2}
 $\rightarrow$ 4 $\rightarrow$ {2, 1} {5, 3}
 $\rightarrow$ 5 $\rightarrow$ {2, 6} {3, 2} {4, 3}

$[(E+V) \log V]$ X

priority-queue
min-heap

distance array (INT-MAX)